Mean Value – Part 2

Questions 1–18

Question 1. Find the exact mean value of each of the following on the given intervals:

a) $f(x) = \pi x \cos(\pi x^2)$ on the interval $\left[0, \frac{\sqrt{2}}{2}\right]$

b) $f(x) = \operatorname{cosec}^2 x$ on the interval $\left[\frac{\pi}{4}, \frac{\pi}{3}\right]$

c) $f(x) = \dfrac{x^2}{5x^3 + 1}$ on the interval $[0, 1]$

d) $f(x) = \sin^2 4x$ on the interval $\left[0, \frac{\pi}{4}\right]$

e) $f(x) = 2^x$ on the interval $[2, 5]$

f) $f(x) = \sin 2x \cos^3 2x$ on the interval $\left[0, \frac{\pi}{12}\right]$

Question 2. Find the mean value of the function $f(x) = \dfrac{x}{e^{-3x}}$ in the interval $0 \leq x \leq a$, giving your answer in terms of a.

Question 3. The mean value of the function $f(x) = \dfrac{2}{\sqrt{3x+1}} + k$ in the interval $[0, 8]$ where k is a positive constant is 2. Find the value of k.

Question 4. The mean value of the function $f(x) = \sqrt{x+1}$ in the interval $[0, 3]$ is half the mean value of the function $g(x) = \frac{3}{7}x^2$ for the interval $[0, k]$ where k is a positive number. Find the exact value of k.

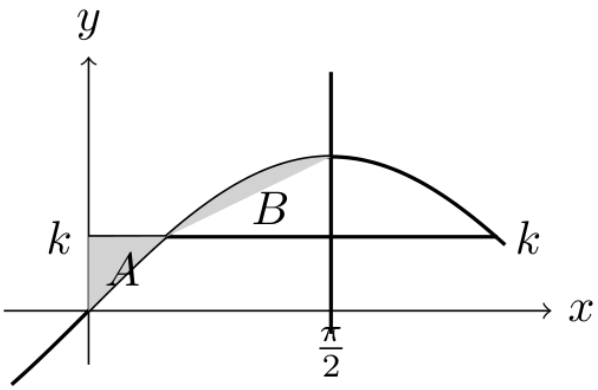
Question 5. The velocity of a car after t seconds is given by $v = \dfrac{30t}{t+5}$ metres per second.

a) Show that the mean velocity for $0 \leq t \leq 20$ is 17.9 m s^{-1} to 3 significant figures.

b) Use the mean value theorem to calculate the mean acceleration of the car for the same time interval.

Question 6. Given that the function f is defined by $f(x) = \ln x$ for $1 \leq x \leq 4$, **a)** Find the mean value of f. Another function g is defined by $g(x) = f(x) + c$ for $1 \leq x \leq 4$. **b)** Given that the mean value of g is $\frac{4}{3}\ln 4$, state the value of c.

Question 7. The diagram below shows the curve $f(x) = \sin x$ with shaded regions A and B enclosed by the y-axis, the line $y = k$ and $x = \frac{\pi}{2}$. The areas A and B are equal.



Determine the value of k.

Question 8. The root mean square (RMS) of a function $f(x)$ defined for $a \leq x \leq b$ over this interval is

$$\sqrt{\frac{1}{b-a}\int_a^b [f(x)]^2\,dx}.$$

Find the RMS of $f(x) = 4\cos x$ from $x = 0$ to $x = 2\pi$ giving your answer in exact form.

Question 9. a) Show that $\cos^4\theta = \frac{1}{8}(3 + 4\cos 2\theta + \cos 4\theta)$. In a mechanical system, the velocity v m s^{-1} of a vibrating component at time t seconds is given by $v = a\cos^2(kt)$, where a and k are positive constants. The effective speed of the component over one complete cycle of $\frac{\pi}{k}$ seconds is defined to be the square root of the mean value of v^2 over this time interval. **b)** Show that the effective speed of the component is $\frac{\sqrt{3}a}{2\sqrt{2}}$ m s^{-1}. **Prerequisite Knowledge for Question 10: Hyperbolic Functions**

Question 10. For each of the following functions, defined over the domain $-k\pi \leq x \leq k\pi$ for positive integer k, find the mean value of $[f(x)]^n$

a) $\sinh(x)$, n is an odd positive integer.
b) $\cosh(ix)$, n is an odd positive integer.
c) $\sinh(ix)$, n is an odd positive integer.

The following questions require the use of A-Level Further Maths Integration techniques

Question 11. Find the exact mean value of each of the following for the given intervals:

a) $f(x) = \dfrac{1}{9+x^2}$ in the interval $[0, 3]$

b) $f(x) = \dfrac{1}{\sqrt{1-4x^2}}$ in the interval $\left[0, \frac{1}{4}\right]$

c) $f(x) = \dfrac{1}{x^2+7}$ in the interval $[0, a]$ for positive constant a

Prerequisite Knowledge for Questions 12–15: Calculus of Hyperbolic Functions

Question 12. Find the exact mean value of the function $f(x) = \dfrac{1}{\sqrt{x^2-6x+16}}$ between $x = 0$ and $x = 6$.

Question 13. Find the mean value of the function $f(x) = \dfrac{1}{\sqrt{x^2 - 9}}$ in the interval $3 \leq x \leq a$, giving your answer in exact logarithmic form.

Question 14. The intensity of a signal, I watts per square metre, at time t hours after a transmitter is switched on is modelled by

$$I = 12 + 8\,\text{sech}^2(0.5t)$$

for $t \geq 0$. Determine the mean signal intensity during the first four hours after the transmitter is switched on, giving your answer correct to three significant figures.

Question 15. The brightness of a light-emitting diode, B in lumens, t minutes after it is switched on, is modelled by

$$B = 50 + 20\,\text{cosech}^2(0.2t)$$

for $t > 0$. Determine the mean brightness of the LED during the second minute after it is switched on, giving your answer to three significant figures.

Question 16. The velocity of a cyclist after t seconds is given by

$$v = \frac{18}{\sqrt{t^2 + 9}}$$

metres per second for $t \geq 0$.

a) Show that the mean velocity of the cyclist for $0 \leq t \leq 12$ is 3.14 m s^{-1} to three significant figures.

b) Use the mean value theorem to calculate the mean acceleration of the cyclist for the same time interval giving your answer to three significant figures.

Question 17. Given that $\cos\theta = \frac{a}{5}$, find the mean value of the function

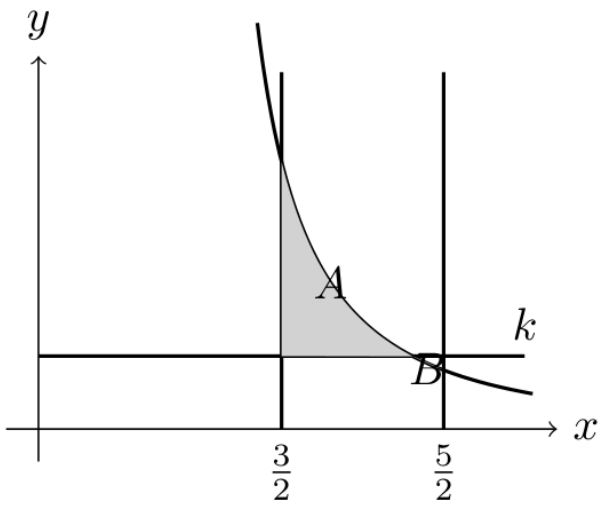
$$f(x) = \frac{1}{\sqrt{25 - x^2}}$$

in the interval $0 \leq x \leq a$, giving your answer in the form $p + q\theta$ where p and q are constants to be found in terms of a. **Prerequisite Knowledge for Question 18: A-Level Further Maths Partial Fractions**

Question 18. The diagram below shows part of the curve

$$f(x) = \frac{x + 2}{(x^2 + 2)(x - 1)}$$

with shaded regions A and B enclosed by the line $y = k$ and the lines $x = \frac{3}{2}$ and $x = \frac{5}{2}$.



The areas A and B are equal. Determine the exact value of k.